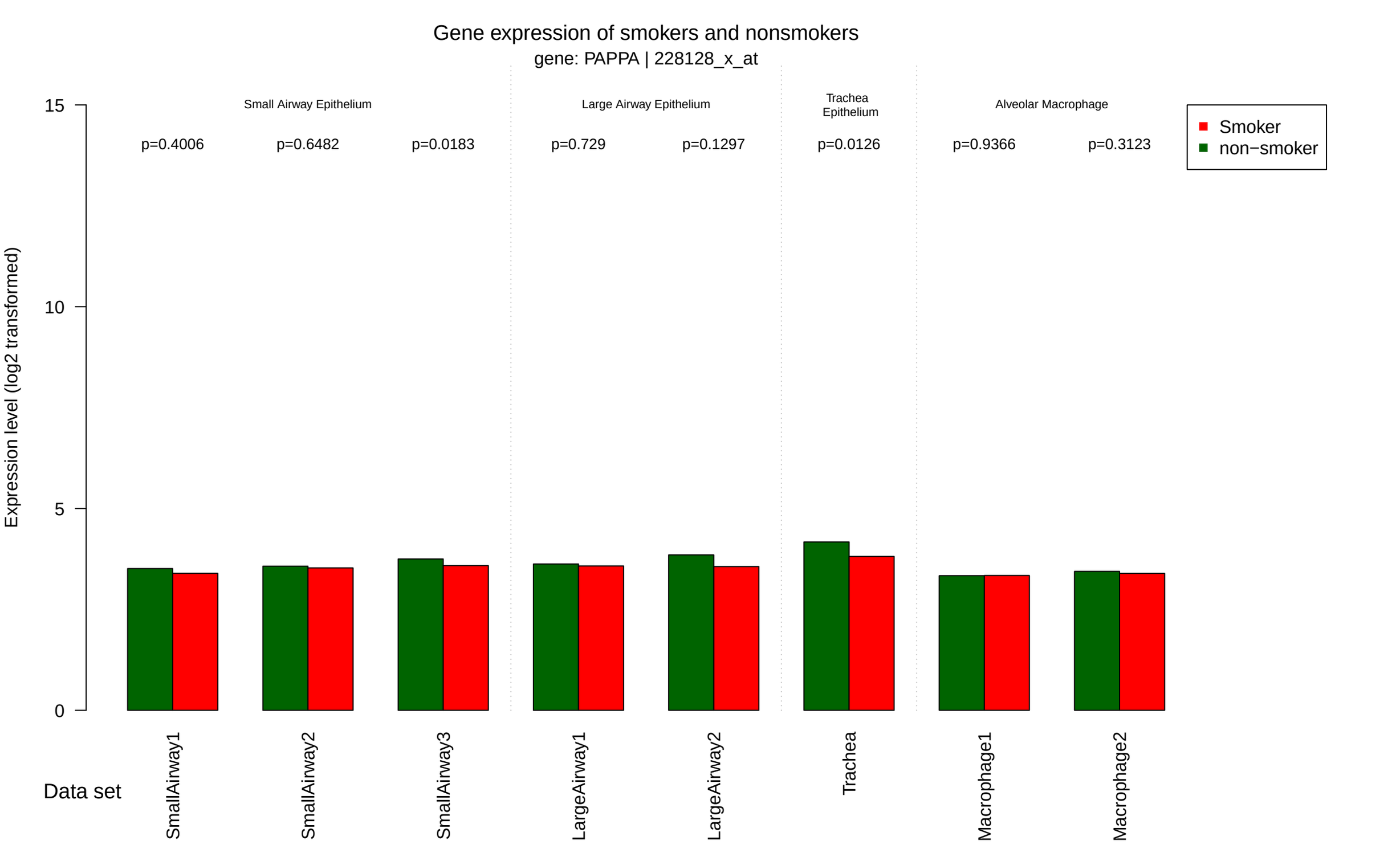
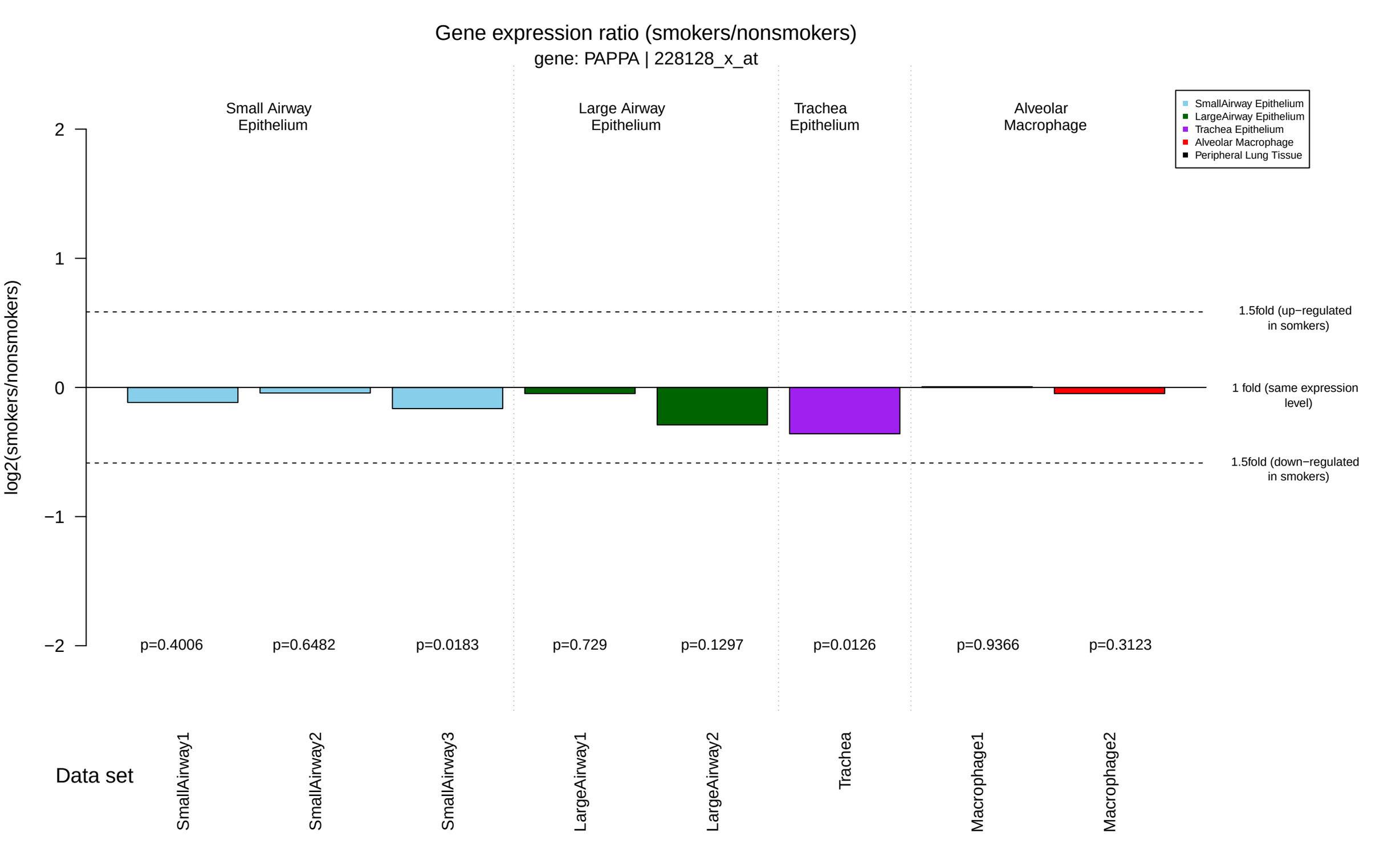
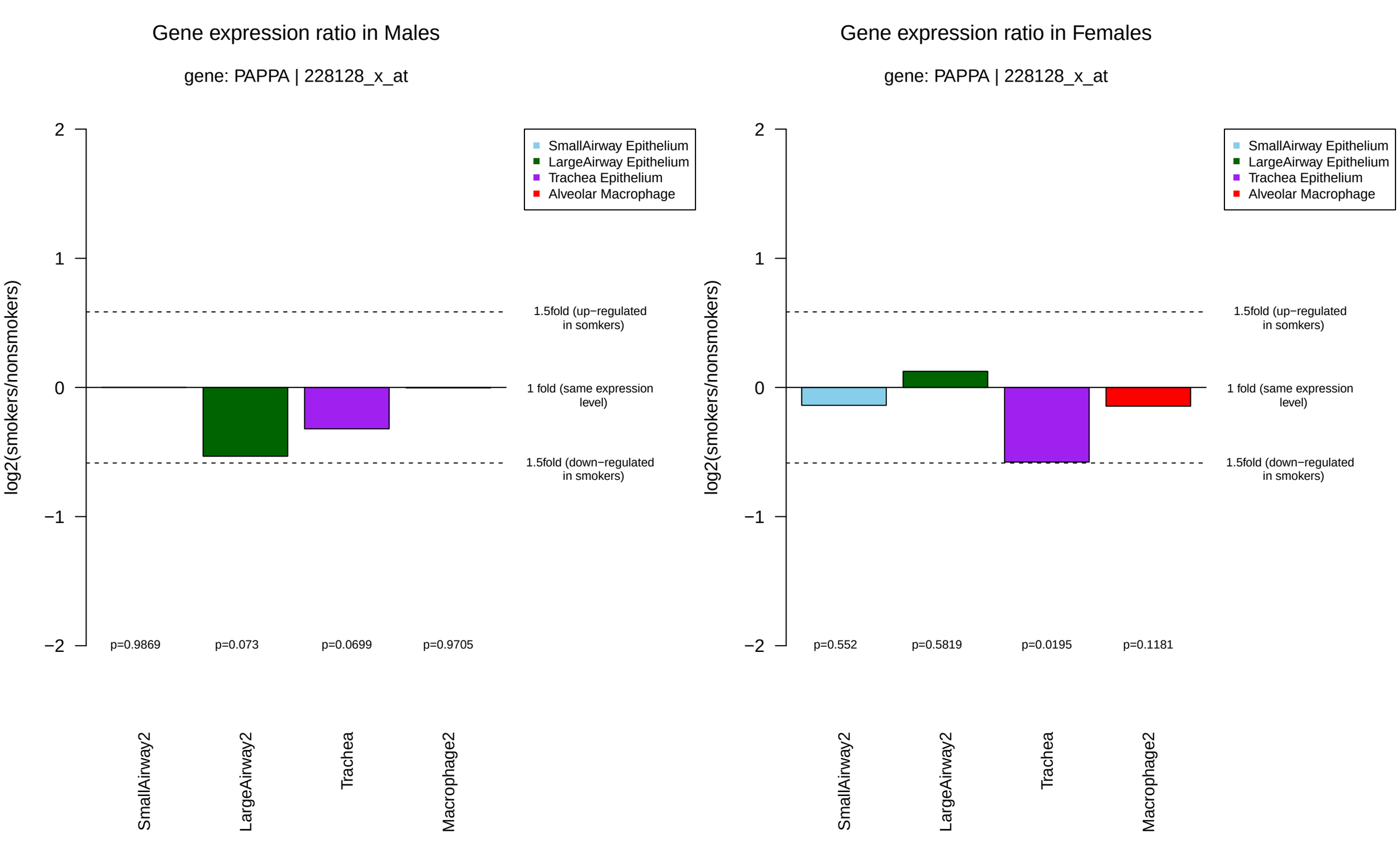
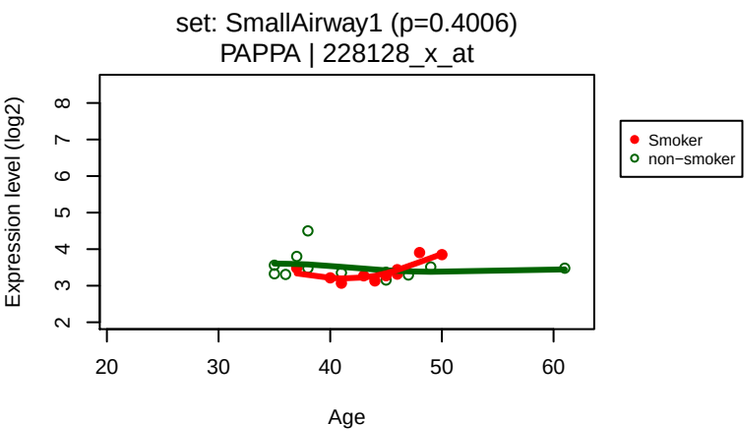
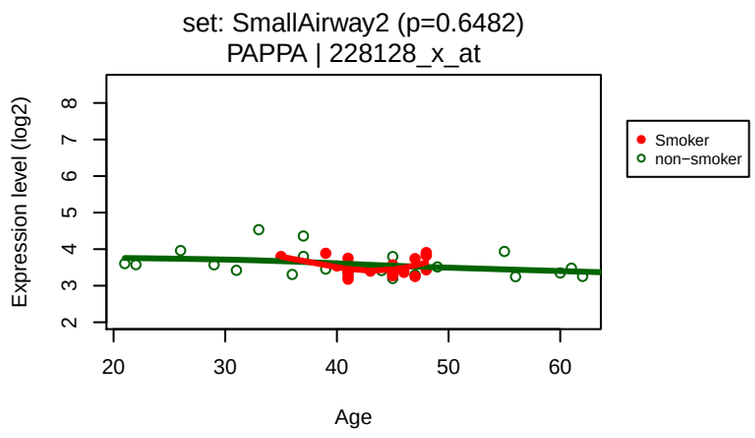
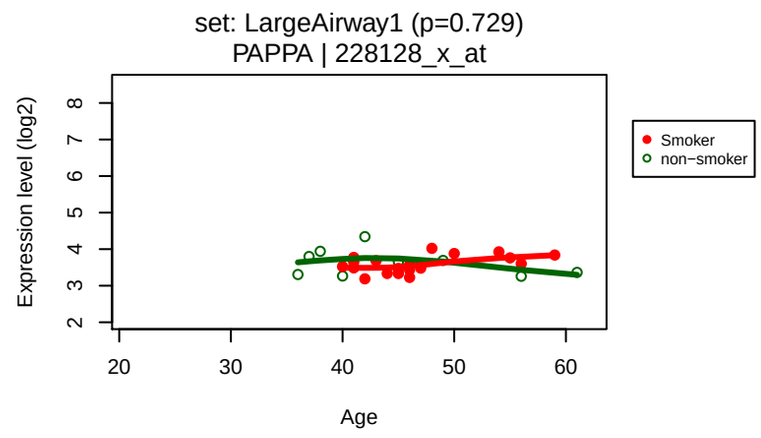

Gene expression of smokers and nonsmokers
gene: PAPPA | 228128_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4006
p=0.6482
p=0.0183
p=0.729
p=0.1297
p=0.0126
p=0.9366
p=0.3123
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

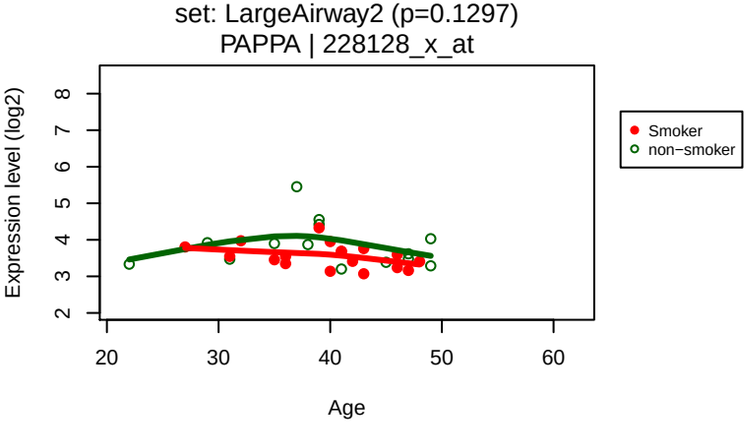

Gene expression ratio (smokers/nonsmokers)
gene: PAPPA | 228128_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4006
p=0.6482
p=0.0183
p=0.729
p=0.1297
p=0.0126
p=0.9366
p=0.3123
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

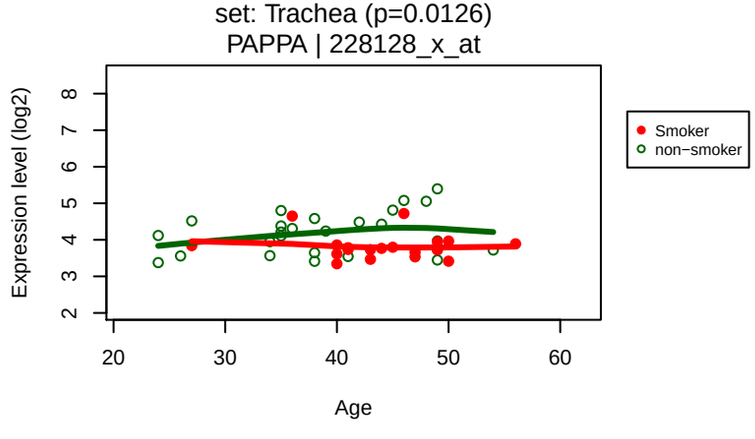

Gene expression ratio in Males
gene: PAPPA | 228128_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9869
p=0.073
p=0.0699
p=0.9705
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPPA | 228128_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.552
p=0.5819
p=0.0195
p=0.1181
SmallAirway2
LargeAirway2
Trachea
Macrophage2

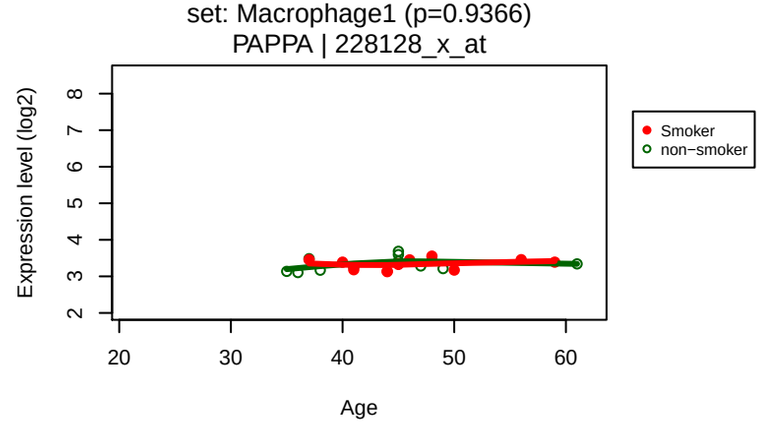

set: SmallAirway1 (p=0.4006)
PAPPA | 228128_x_at
Expression level (log2)
Age
Smoker
non−smoker

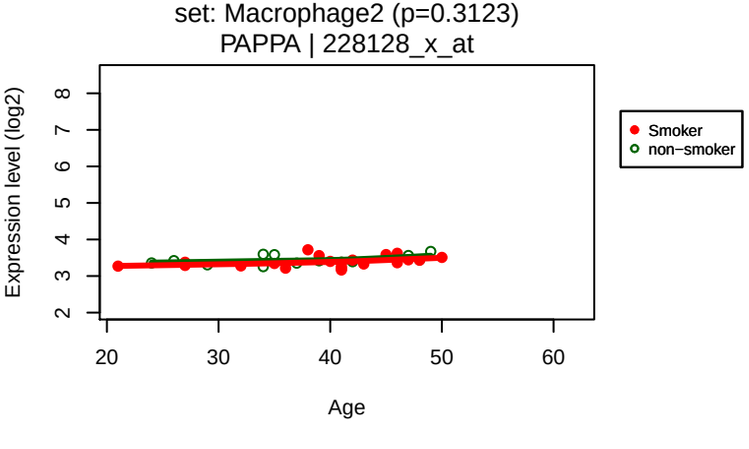

set: SmallAirway2 (p=0.6482)
PAPPA | 228128_x_at
Expression level (log2)
Age
Smoker
non−smoker

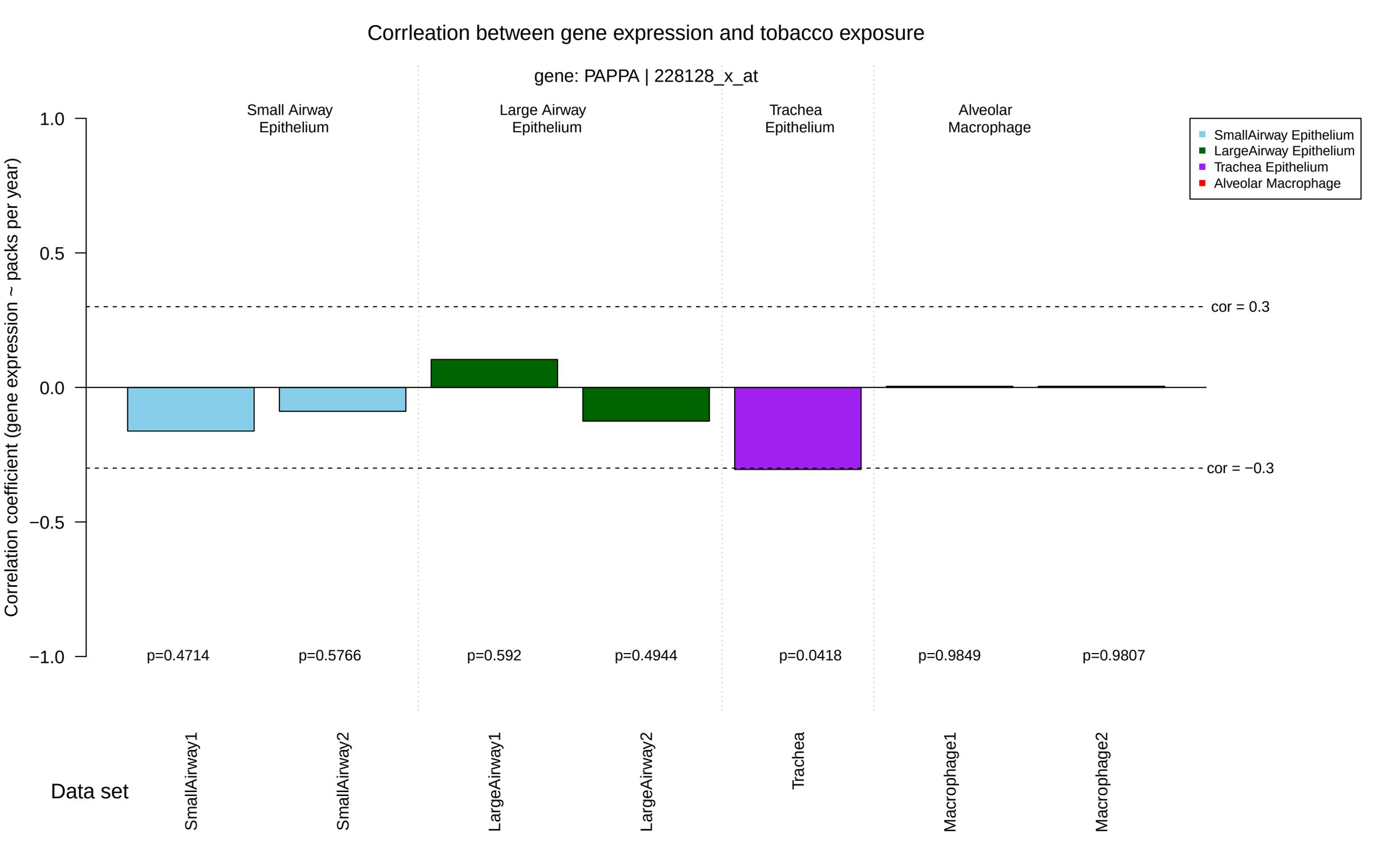

set: LargeAirway1 (p=0.729)
PAPPA | 228128_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1297)
PAPPA | 228128_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0126)
PAPPA | 228128_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9366)
PAPPA | 228128_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3123)
PAPPA | 228128_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPPA | 228128_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4714
p=0.5766
p=0.592
p=0.4944
p=0.0418
p=0.9849
p=0.9807
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2